

FondoMéxicoAlfa

A Systematic Long-Short Strategy for Mexican Equities and FIBRAs

Multi-provider data infrastructure, Black–Litterman portfolio construction, machine-learning attribution, and macro-regime conditioning

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Abstract

We document the construction and out-of-sample evaluation of a systematic long-short strategy for the joint universe of Mexican publicly traded equities and FIBRAs (the Mexican variant of REITs). The strategy is built on a multi-provider data infrastructure (Bloomberg, Refinitiv, Yahoo Finance), a Black–Litterman portfolio construction layer that combines machine-learning views with low-confidence macro tilts, mean-variance and CVaR optimizers operating under CNBV regulatory constraints, a Layer 2 FX hedge overlay, and a complete suite of overfitting diagnostics including deflated Sharpe ratios and combinatorially symmetric cross-validation. We extend the pipeline with an XGBoost cross-sectional return forecaster, TreeExplainer SHAP attribution, and Banxico macro-regime conditioning. Evaluated on 108 monthly walk-forward out-of-sample rebalances from 2017 through early 2026 with Bloomberg point-in-time fundamentals, the regulated portfolio achieves an annualized Sharpe ratio of 0.44, an annualized return of 8.34% at 13.59% volatility, and a maximum drawdown of –35.5%. ElasticNetCV and XGBoost produce statistically indistinguishable performance on real data; the value of the gradient-boosted model lies in its attribution framework rather than raw return lift. SHAP analysis identifies FIBRA-specific operating metrics — loan-to-value, FFO yield, capitalization rate — as the dominant return predictors, with stability of 0.44 (Spearman) across consecutive rebalances. Regime conditioning reveals materially higher reliability during Banxico easing cycles than during tightening cycles. We argue that the binding constraint on machine-learning approaches to this market is the small effective cross-section of the Mexican universe.

Keywords: *quantitative equity, factor investing, FIBRAs, emerging markets, Black–Litterman, gradient boosting, SHAP, macro regimes, walk-forward validation, CNBV regulation.*

1. Introduction

The systematic investment literature has converged on a relatively uniform methodology for developed equity markets: factor portfolios constructed from large cross-sections, ranked by composite scores spanning value, quality, momentum, and low-volatility signals, and reweighted by mean-variance or risk-parity optimization. The methodology works because the cross-section is large enough — typically several thousand stocks in the US and several hundred in major European markets — for the law of large numbers to overwhelm the noise in individual signals. When the same methodology is applied to emerging markets, results typically degrade in proportion to the cross-section size: regional pooling helps, but the per-country signal is fragile.

The Mexican equity market is an extreme case of this problem. The IPC index contains roughly thirty constituents; the float-adjusted investable universe is somewhat smaller. At the same time, the local fixed-income market is deep, the regulatory environment is sophisticated (CNBV operates a position-concentration regime modeled on European UCITS), and the FIBRA market — Mexico's variant of REITs, introduced by reform in 2004 — has matured into a meaningful component of the listed real-asset universe. Mexico is therefore too small to be approached with standard developed-market methods and too

institutionally complex to be approached as a pure emerging-market discretionary play.

This paper documents a complete systematic framework calibrated to these specific constraints. The contributions are three. First, the framework integrates a set of FIBRA-specific fundamental features — loan-to-value, FFO yield, capitalization rate, vacancy rate — into a unified cross-sectional model alongside equity fundamentals and price-based signals. These features have no direct analog in standard equity factor libraries, and they are constructed point-in-time from Bloomberg fundamentals with a 90-day reporting lag. Second, the framework layers a Black–Litterman posterior over the cross-sectional model output: per-ticker return views from ElasticNetCV or XGBoost are combined with low-confidence (0.20) macro sector views derived from industrial production, exports, the Banxico overnight target rate, the USDMXN exchange rate, and US inflation. The Black–Litterman step is essential in a small cross-section because it controls the variance of the forecast inputs that feed the optimizer; without it, the mean-variance solution becomes unstable across rebalances. Third, the framework includes a complete suite of overfitting and model-reliability diagnostics: SHAP-based feature stability metrics, Banxico regime conditioning, deflated Sharpe ratios, and combinatorially symmetric cross-validation following Bailey and López de Prado (2014, 2016). The diagnostics are not appendices to the main result; they are the result, in the sense that they tell the operator when the model is trustworthy and when it is not.

The paper proceeds as follows. Section 2 describes the data infrastructure and universe. Section 3 details the feature engineering. Section 4 lays out the modeling methodology. Section 5 presents results. Section 6 discusses what we view as the most important methodological finding of the work — that XGBoost does not materially outperform a well-regularized linear baseline in this universe — and the practical implications for live deployment. Section 7 concludes.

2. Data Infrastructure and Universe

The data layer integrates four providers with automatic fallback. Bloomberg via BLPAPI is the production source, providing point-in-time fundamental data with full historical coverage from 2017 through early 2026. Refinitiv/LSEG is an institutional alternative with somewhat shorter coverage for several constituents. Yahoo Finance is available without credentials but provides only a snapshot of current fundamentals; for the backtest window, only price-based signals can be constructed from this source. Macro data — Banxico overnight target rate, TIIE 28-day rate, USDMXN, US industrial production, US inflation — is sourced from Banxico SIE and FRED.

The pipeline operates under a *strict_data_mode* flag set to true in the production configuration. When this flag is enabled, the pipeline aborts on any data failure rather than substituting synthetic or default values. This is the appropriate behavior for a research and pre-production system where silent data substitution is worse than an explicit failure. All fundamental data is lagged by 90 calendar days to approximate the information set available to a real-time investor; restated historical values that would not have been available in real time are excluded.

The investable universe consists of Mexican equities listed on the Bolsa Mexicana de Valores (BMV) and FIBRAs traded on the same exchange. The cross-section comprises approximately twenty-six equities and six FIBRAs at each rebalance date, with modest variation due to corporate actions. The backtest period is monthly rebalances from January 2017 through March 2026 — 108 rebalance dates after warm-up. Returns are total-return, inclusive of dividends and distributions, expressed in Mexican pesos.

Several CNBV regulatory constraints are imposed at every rebalance. Individual positions are capped at 10% of NAV, with a consolidated 10% issuer concentration limit applied across tickers sharing the same issuer ID. A liquidity sleeve of CETES28 and CETES91 instruments occupies 3% to 15% of NAV depending on the prevailing macroeconomic regime, expanding in stress periods. These constraints are fixed structural parameters, not variables in the hyperparameter search.

3. Feature Engineering

Three categories of features feed the cross-sectional model. Equity fundamentals include the price-to-earnings ratio, return on equity, EBITDA growth, profit margin, net debt to EBITDA, capital expenditure to sales, and dividend yield. These are the standard set of value, quality, and capital-allocation factors that the global factor literature has documented (Asness, Frazzini, and Pedersen, 2019; Fama and French, 2015).

FIBRA-specific fundamentals are the central methodological contribution of the feature layer. The FIBRA structure produces a set of operating metrics that have no direct analog in equity reporting and that are routinely used by real-asset specialists to assess relative value. We include four: the capitalization rate (net operating income divided by enterprise value, the FIBRA analog of an earnings yield but constructed from property-level cash flow rather than accounting earnings); FFO yield (funds from operations divided by market capitalization, the standard REIT-industry valuation metric); loan-to-value (total debt divided by gross asset value, a measure of leverage that for property portfolios has substantially more predictive power for distress risk than the corresponding accounting leverage measures because property collateral is more directly marked to market); and vacancy rate (the percentage of leasable area not currently under contract, a forward-looking operating metric that has no equivalent in conventional equity reporting).

For equity securities, the four FIBRA-specific features are set to the cross-sectional median (effectively, a neutral exposure). For FIBRAs, equity-specific features that lack operating meaning are likewise neutralized. This treatment lets the model identify, empirically, whether the FIBRA features carry pricing information that the equity features cannot replicate.

Cross-sectional technical features are limited to a 63-trading-day momentum signal and a 63-trading-day realized volatility signal. Short-horizon momentum has well-documented characteristics in emerging-market equities (Cakici, Fabozzi, and Tan, 2013). All features are standardized cross-sectionally within each rebalance date (z-score against the contemporaneous cross-section) before entering the model. Cross-sectional rather than time-series standardization is the appropriate choice for a relative-value framework: the model is asked to forecast which assets will outperform their peers, not whether the overall market will rise or fall.

4. Methodology

4.1 Walk-forward validation

The pipeline is trained and evaluated under strict walk-forward out-of-sample discipline. At each monthly rebalance date t , the cross-sectional model is fitted on data available through $t-1$, used to generate forecasts for t , and the forecasts are evaluated against realized returns from t to $t+1$. The training window expands monthly; we do not use a rolling window because the panel is short enough that discarding old observations would meaningfully reduce statistical power. Hyperparameter selection for the gradient-boosted model uses an inner time-series cross-validation that operates only on the training window, with five expanding-window splits and early stopping on the final fold's holdout. Nested validation discipline of this kind is essential: shortcuts such as a single random validation fold or hyperparameter optimization over the full sample produce performance estimates that are not realizable in live trading (López de Prado, 2018).

4.2 ElasticNetCV baseline and XGBoost alternative

The baseline cross-sectional model is an elastic-net linear regression with hyperparameter selection by cross-validation. The L1/L2 mixing parameter and regularization strength are selected by the inner time-series CV. The elastic net is the appropriate linear baseline for this problem: it handles the collinearity among fundamental features and the small sample size more gracefully than either OLS or pure ridge and lasso variants. The alternative is XGBoost (Chen and Guestrin, 2016) configured as a regressor with

mean-squared-error loss, wrapping an internal RandomizedSearchCV over a hyperparameter space spanning tree depth, learning rate, subsample and feature-subsample ratios, L1 and L2 leaf regularization, and minimum child weight. The number of trees is selected dynamically by early stopping with a hard cap of 2000 boosting rounds.

4.3 Black–Litterman posterior

Forecasts from the cross-sectional model do not feed the portfolio optimizer directly. They are instead converted into per-ticker views and combined with macro sector views in a Black–Litterman posterior (Black and Litterman, 1992). Per-ticker views are assigned confidence weights derived from the model's in-sample fit quality. Macro sector views — derived from industrial production, exports, the Banxico target rate, USDMXN momentum, and US inflation — are blended at low confidence (0.20) specifically chosen so that macro information nudges rather than dominates the quantitative signal. The Black–Litterman step is essential in a small cross-section: raw cross-sectional forecasts have substantial variance across rebalances, and feeding them directly into a mean-variance optimizer produces a portfolio that turns over excessively. The BL posterior pulls the expected-return vector toward the equilibrium prior weighted by view confidence, which materially stabilizes the optimizer output without dampening the underlying signal.

4.4 Portfolio optimization

The expected-return vector from the BL posterior enters a portfolio optimizer that operates under CNBV regulatory constraints. Three solvers are available: a sequential least-squares mean-variance optimizer with a market-impact penalty, a min-CVaR optimizer at 95%, and a Michaud robust solver that averages 100 mean-variance solutions with bootstrap-perturbed expected returns. The covariance matrix is estimated by EWMA with Ledoit-Wolf shrinkage toward the constant-correlation target (Ledoit and Wolf, 2004). Transaction costs are applied at 10 basis points per side on the change in weights at each rebalance. The regulatory NAV reported in this paper uses the mean-variance solution.

4.5 Macro-regime classification and SHAP attribution

We classify each rebalance into one of three Banxico rate regimes (TIGHTENING, EASING, NEUTRAL based on the trailing three-month change in the target rate) and one of two market stress regimes (STRESS, CALM based on the 60-day IPC realized-volatility quartile). Regime labels at time t use only data available at $t-1$. After the XGBoost model is fitted at each rebalance, we construct a TreeExplainer SHAP instance from the fitted estimator (Lundberg et al., 2020) and compute SHAP values for the test slice. These accumulate across rebalances into a panel indexed by (date, ticker, feature, shap_value). The mean absolute SHAP value per feature at each rebalance produces a ranking; the Spearman rank correlation of this ranking between consecutive rebalances is our stability metric. A stability of 0.80 or higher is typically required for institutional deployment of a tree-based signal.

5. Results

5.1 Performance: ElasticNetCV and XGBoost on real data

Table 1 reports the headline out-of-sample performance metrics for both models on each of the three data sources. The figures are for the regulated NAV portfolio under the mean-variance optimizer with CNBV constraints applied; transaction costs are 10 basis points per side, hedge overlay excluded.

Table 1. Out-of-sample performance, regulated NAV, January 2017 – March 2026.

Source	Model	Return	Vol	Sharpe	Sortino	Max DD	CVaR 95%	Turnover
Bloomberg	ElasticNetCV	8.34%	13.59%	0.44	0.45	–35.50%	–1.95%	0.57%
Bloomberg	XGBoost	8.29%	13.57%	0.44	0.44	–35.38%	–1.95%	6.33%

Yahoo	ElasticNetCV	9.98%	15.86%	0.47	0.48	−36.58%	−2.26%	0.04%
Yahoo	XGBoost	9.89%	15.81%	0.47	0.48	−36.58%	−2.25%	0.44%
Refinitiv	ElasticNetCV	5.80%	16.01%	0.23	0.23	−43.19%	−2.28%	0.04%
Refinitiv	XGBoost	5.73%	15.60%	0.23	0.23	−42.75%	−2.23%	0.67%

All figures are walk-forward OOS. Hedge overlay excluded. Bloomberg uses point-in-time fundamentals; Yahoo uses price signals only (no historical fundamentals available).

Two findings stand out. First, within each data source, ElasticNetCV and XGBoost produce statistically indistinguishable performance. The Sharpe ratios are identical to two decimal places across all three providers; returns differ by less than five basis points annualized; drawdowns differ by less than one percentage point. The 95% confidence interval on the Bloomberg Sharpe estimate (paired stationary bootstrap, 5000 replications) is [−0.25, 1.18], which dwarfs any difference between the two models. There is no meaningful sense in which XGBoost outperforms the linear baseline on this data. The only material difference between the two models is turnover, where XGBoost is approximately ten times higher than ElasticNet — a direct cost driver that, absent a corresponding return advantage, is a liability rather than an asset.

Second, performance differs substantially across data providers. Bloomberg achieves Sharpe 0.44 with point-in-time fundamentals and full historical coverage; Yahoo Finance achieves marginally higher Sharpe (0.47) but using only price signals, because historical fundamental data is not available through that channel; Refinitiv falls to Sharpe 0.23, attributable to coverage gaps in the local-equity feed that compress cross-sectional dispersion. The Yahoo result is not directly comparable to Bloomberg because the signal sets differ — the counterintuitive finding that a smaller signal set produces a higher Sharpe in this sample reflects the contribution of model-fit variance from the additional fundamental features in a small cross-section, not a real informational advantage of price-only signals.

5.2 SHAP feature attribution

Despite the absence of a performance lift from XGBoost, the SHAP attribution framework produces interpretable and economically sensible results. Table 2 reports the top ten features by time-averaged mean absolute SHAP value across the Bloomberg walk-forward sample.

Table 2. Top features by time-averaged mean |SHAP| value.

Rank	Feature	Mean SHAP	Std SHAP
1	ltv	0.00350	0.00363
2	ffo_yield	0.00279	0.00360
3	cap_rate	0.00221	0.00236
4	pe_ratio	0.00196	0.00233
5	dividend_yield	0.00183	0.00214
6	momentum_63	0.00172	0.00249
7	roe	0.00154	0.00183
8	ebitda_growth	0.00146	0.00188
9	profit_margin	0.00145	0.00220
10	capex_to_sales	0.00136	0.00220

Computed across 107 walk-forward rebalances. FIBRA-specific features occupy positions 1–3.

Three of the four FIBRA-specific features occupy the top three positions, with loan-to-value alone showing a mean |SHAP| value 1.8 times larger than the highest equity feature (price-to-earnings). The economic interpretation is that FIBRA pricing is driven by operating metrics — property-level cash flow yields, leverage measured against marked-to-market collateral — that have no accounting analog in the equity feature set. A factor model built exclusively on equity-style features would systematically misprice FIBRAs; combining the two universes is informationally productive precisely because the FIBRA features add genuinely orthogonal signal.

Table 3. SHAP feature-rank stability across consecutive rebalances.

K	Pairs	Mean Spearman	Std Spearman
Top 5	107	0.440	0.421
Top 10	107	0.428	0.329
All	107	0.455	0.292

Spearman rank correlation between consecutive rebalances of the mean |SHAP| ranking, restricted to the top-K features.

The mean stability of 0.44 is well below the 0.80 threshold typically required for production deployment. We attribute the instability to the small effective cross-section. With approximately thirty assets in the universe, each rebalance's training data provides limited statistical power to pin down feature importance, and the stochastic component of the boosting algorithm amplifies the resulting noise. Gu, Kelly, and Xiu (2020) document that tree ensembles deliver meaningful performance lift over linear baselines only when the cross-section is large enough that within-period information dominates the noise in feature attribution; the Mexican universe is on the wrong side of that threshold.

5.3 Regime-conditioned performance

Table 4 reports the regime-conditioned metrics, aggregated across stress regimes to focus on the Banxico rate-regime dimension.

Table 4. Performance metrics conditional on Banxico rate regime.

Rate regime	N	IC mean	ICIR	SHAP stab (top-5)	Momentum SHAP
TIGHTENING	64	+0.125	0.71	0.41	−0.000136
EASING	33	+0.130	0.94	0.57	+0.000110
NEUTRAL	11	+0.136	—	0.39	−0.000934

Momentum SHAP is the signed (not absolute) mean SHAP contribution of momentum_63 in each regime.

The model is meaningfully more reliable in easing cycles than in tightening cycles. SHAP stability is 0.57 in EASING versus 0.41 in TIGHTENING — neither value reaches the production threshold, but the gap is large enough to warrant operational attention. The ICIR is correspondingly higher in EASING (0.94 versus 0.71). The economic intuition is consistent with standard emerging-market equity literature: falling discount rates amplify cross-sectional dispersion among assets with differing duration profiles and operating leverage, and the dispersion is what the cross-sectional model exploits. In EASING regimes, momentum_63 enters with a small positive SHAP contribution, consistent with trend continuation when monetary policy is supportive. In TIGHTENING, the sign flips negative, consistent with the reversal pattern documented in the emerging-market literature.

5.4 Hyperparameter optimization and overfitting diagnostics

The Bayesian search over the broader pipeline parameter space (50 trials per source, purged walk-forward cross-validation with 21-day gap) produces best validated Sharpe estimates of 0.43 on Bloomberg, 0.57 on Yahoo, and 0.26 on Refinitiv. The deflated Sharpe ratio adjusts the observed Sharpe for skewness, kurtosis,

and the number of configurations tested (Bailey and López de Prado, 2014). Under the null hypothesis of zero skill, the expected maximum Sharpe across 50 trials is 2.23 on the Bloomberg sample. The observed best validated Sharpe of 0.43 is below this ceiling, which is the appropriate honest finding: the search does not produce evidence of statistically distinguishable skill against the multiple-testing-adjusted null. These results are not a failure of the search; they are a calibration of expectations. A Sharpe of 0.43 on a CNBV-regulated, low-turnover, transaction-cost-net portfolio in a small emerging-market universe is a defensible outcome.

5.5 Layer 2 analytical overlay

For completeness, the Bloomberg run with the Layer 2 FX hedge overlay activated achieves an annualized return of 45.7%, an annualized volatility of 23.7%, and a Sharpe ratio of 1.50 on an analytical basis. These figures are not included in the regulated NAV because the Layer 2 overlay operates with leverage and currency positions that fall outside the CNBV-reportable scope; they are reported as a reference for what the same signal stack could achieve under a less constrained mandate. The LFI reform scenario analysis compares the regulated structure against three alternatives; the 130/30 structure on Bloomberg achieves a Sharpe of 1.82 versus 1.47 for the regulated structure on the hedge basis.

6. Discussion

The most important methodological finding of this work is that the XGBoost cross-sectional forecaster does not materially outperform the elastic-net baseline on real Mexican-market data. This is not a failure of the implementation — both models are constructed with internal walk-forward cross-validation, both are fitted under identical out-of-sample discipline, both are evaluated on the same rebalance sequence — and it is not a surprising result given the constraints of the universe. It is, instead, a useful negative finding that has direct implications for live deployment and for the broader literature on machine learning in emerging-market equities.

The mechanism is the small effective cross-section. Tree ensembles draw their predictive advantage over linear models from the ability to learn non-linear interactions among features without overfitting, but this advantage requires sufficient within-period information for the algorithm to identify those interactions reliably. With roughly thirty assets at each rebalance, the available information per training window is below the threshold at which the algorithm can extract reliable non-linear structure; the model defaults toward the same relationships that the linear baseline already captures, plus added noise from the boosting randomization. The 0.44 SHAP feature-rank stability is a direct measurement of this noise. The 10× turnover differential without a corresponding return advantage is its portfolio-level manifestation.

This finding does not mean the XGBoost component is without value. The attribution framework — SHAP per rebalance, feature-rank stability, regime-conditioned performance — is genuinely informative independent of whether XGBoost itself generates excess return. The framework tells the operator which features the model is weighting, how those weights move over time, and under what macroeconomic conditions the model is reliable. These are operational diagnostics that the linear baseline cannot produce, and they have value even when the linear baseline matches the gradient-boosted model on raw performance.

Three operational responses follow from these findings. The first is regime-conditional model selection: use the XGBoost forecasts during EASING regimes, where SHAP stability reaches 0.57, and revert to the ElasticNet baseline during TIGHTENING and NEUTRAL regimes. The regime classification is itself a one-period-lagged signal with no lookahead risk. The second is feature pruning: the SHAP decomposition identifies momentum_63 as a substantial contributor to turnover without proportionate contribution to predictive accuracy. The third is ensemble blending: combining ElasticNet and XGBoost predictions at weights inversely proportional to their out-of-sample forecast variance would inherit the stability of the linear baseline and add whatever marginal lift the gradient-boosted model can deliver in favorable regimes.

Several limitations deserve explicit acknowledgment. The strategy is calibrated specifically to the Mexican universe and would not transfer to other emerging markets without recalibration, particularly because the FIBRA-specific features depend on the existence of a deep, liquid REIT segment within the local market. A more rigorous treatment of transaction costs would model market impact as a square-root function of trade size relative to daily volume, which would have a disproportionate effect on the high-turnover XGBoost configuration. Finally, the 95% confidence interval on the Sharpe estimate is wide because the nine-year backtest contains only one full Banxico tightening cycle and one easing cycle.

7. Conclusion

We have documented a complete systematic framework for the Mexican equity and FIBRA universe, evaluated under strict walk-forward out-of-sample discipline across three data providers. The framework integrates multi-provider data infrastructure, a Black–Litterman portfolio construction layer that blends machine-learning views with low-confidence macro tilts, multiple optimizers operating under CNBV regulatory constraints, a Layer 2 FX hedge overlay reported on an analytical basis, an XGBoost cross-sectional forecaster with TreeExplainer SHAP attribution, Banxico macro-regime conditioning, and a complete suite of overfitting diagnostics. The regulated portfolio achieves a Sharpe ratio of 0.44 on Bloomberg point-in-time fundamentals over the 2017–2026 window.

The principal empirical findings are three. First, FIBRA-specific operating metrics — loan-to-value, FFO yield, capitalization rate — dominate the SHAP attribution of the gradient-boosted model and carry pricing information that conventional equity factors cannot replicate. Second, XGBoost and ElasticNetCV produce statistically indistinguishable performance on real data within the Mexican cross-section, attributable to the small effective universe; the value of the gradient-boosted model lies in its attribution framework rather than in raw return lift. Third, model reliability is regime-dependent: Banxico easing cycles produce SHAP feature-rank stability of 0.57 against 0.41 in tightening cycles, with corresponding differences in the information coefficient.

The work points toward three directions for further research: out-of-region replication of the framework on the Brazilian Fundos Imobiliários market, more granular macroeconomic conditioning beyond the binary regime classification, and the construction of an ensemble between the ElasticNet and XGBoost paths with regime-conditional weights that exploits the asymmetric reliability of the two models across the macroeconomic cycle. The framework as constructed provides the diagnostic infrastructure to make such an ensemble robust; the empirical work of fitting it remains to be done.

The strategy in its current form is a research prototype, not a production system. The most useful artifact of this work is not the reported Sharpe ratio but the framework itself: an honest, instrumented, regime-aware approach to a market where naive applications of factor investing and machine learning both fail for diagnosable reasons.

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Appendix A. Hyperparameter Search Space

The XGBoost internal RandomizedSearchCV samples 20 configurations from the following distributions for each training window, with a 5-fold expanding-window TimeSeriesSplit and early stopping (50 rounds).

Parameter	Distribution
max_depth	uniform integer {3, 4, 5, 6}
learning_rate	uniform {0.01, 0.03, 0.05, 0.10}
subsample	uniform {0.7, 0.8, 1.0}
colsample_bytree	uniform {0.7, 0.8, 1.0}
min_child_weight	uniform {1, 5, 10}
reg_alpha	uniform {0, 0.1, 1.0}
reg_lambda	uniform {0.1, 1.0, 10.0}
n_estimators	early-stopped (cap 2000)

Appendix B. Software and Reproducibility

All results are reproducible from the open-source repository accompanying this paper. The software stack is Python 3.10 or higher with `xgboost` ≥ 2.0 , `shap` ≥ 0.45 , `scikit-learn`, `pandas`, `numpy`, `Optuna`, and `CVXPY` (CVaR optimization). Random seeds are fixed across all stochastic components. The test suite comprises 107 unit and integration tests. Repository: github.com/MaxHidalgoLeon/FondoMexicoAlfa